

Medha Tripathi

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Professional Summary

Software Engineer with a strong foundation in programming, web development, cloud technologies, and software deployment. Experienced in building practical applications, integrating APIs, managing databases, and solving technical problems across frontend and backend systems.

EDUCATION

University of Texas Rio Grande Valley – <i>Edinburg, Texas</i>	May 2025
M.S. Computer Science	
Vellore Institute of Technology – <i>Vellore, India</i>	June 2023
B.Tech Computer Science and Engineering	

PROJECTS

Manobal | *Python / Cryptography / Docker*

- Developed a locally hosted password manager in Python to safely encrypt, store, and manage user credentials.
- Implemented custom local encryption using Python cryptography library Fernet, ensuring data privacy without relying on external servers and creating a secure system.
- Containerized the application using Docker to enable seamless cross-platform deployment.

location2time | *Python / API Integration*

- Engineered a location-to-time software in Python that dynamically computes and returns the accurate local time based on location inputs.
- Architected an algorithm that uses RESTful API integrations to convert input geographical locations into precise coordinates to generate localized date and time objects

Search Engine | *Python / Web Scraping*

- Developed a local search engine utilizing BeautifulSoup to parse, scrape, and extract text data from a large corpus of interlinked HTML pages.
- Implemented a text retrieval process by using data structures like queues, hash maps, and an inverted index.
- Designed and deployed a lightweight web application using Flask to provide an interface for executing search queries and viewing matching results.

EXPERIENCE

Technical Tutor <i>Varsity Tutors, Remote</i>	10/2023-05/2026
- Mentored a broad client base of over 120 graduate and high school students in computer science fundamentals and programming, guiding them through logic building, algorithm design, and system debugging. - Translated complex concepts into easy-to-understand lessons, adapting to diverse communication styles. - Developed customized coding exercises to build practical software development skills and computer literacy.	

Intern Software Engineer <i>Aeromatrix, Uttar Pradesh, India</i>	01/2021-03/2022
- Collaborated on the frontend using HTML and CSS, and the backend in PHP of the e-commerce platform for the startup. - Hosted website on AWS and managed company's inventory database, handling over 250 listings daily. - Integrated backend PHP application with MySQL for persistent data storage.	

TECHNICAL SKILLS

- Languages: Python, Java, C, Ruby, PHP, HTML, CSS, MySQL, MATLAB, Rust, Go, JavaScript
- Frameworks: Flask, Django, FastAPI, React, PyTorch, Angular, Ruby on Rails
- Tools: Git, VS Code, VirtualBox, Figma, Docker, AWS EC2, Linux, Google Cloud Computing, LLM Agents
- Soft Skills: Customer service, technical writing, clear communication, time management